The Callaway and Sant'Anna (2021) and Sun and Abraham (2021) estimators

For readers who know two-way fixed effects (TWFE) regression. CS: Callaway and Sant'Anna; SA: Sun and Abraham.

1. The problem

Unit i starts treatment in period G_i and stays treated ($G_i = \infty$ if never treated); cohort g is the set of units with $G_i = g$, and $\ell = t - g$ is event time. The TWFE event-study regression is

$$Y_{it} = \alpha_i + \lambda_t + \textstyle\sum_{\ell \neq -1} \mu_\ell \mathbf{1}\{t - G_i = \ell\} + \varepsilon_{it}.$$

With staggered adoption and effects that differ across cohorts, μ_ℓ is a linear combination of cohort effects at event time ℓ and at other event times, so under parallel trends and no anticipation a lead ($\ell < 0$) can differ from zero (SA). Goodman-Bacon (2021) shows that the static TWFE coefficient is a weighted average of all two-group, two-period difference-in-differences (DiD) estimates, including comparisons with an already-treated control cohort.

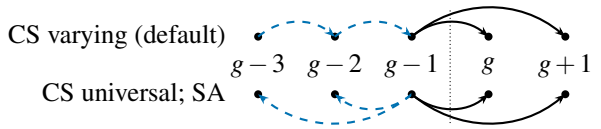


Figure 1: Base periods for cohort g. An arrow from b to t marks the estimate for t with base period b. Blue dashed: pre-treatment; black: post-treatment; dotted: treatment starts.

2. Callaway and Sant'Anna

CS define $\mathrm{ATT}(g,t) = E[Y_t(g) - Y_t(0) \mid G = g]$, the average effect in period t for cohort g, where $Y_t(g)$ and $Y_t(0)$ are potential outcomes under first treatment in g and under no treatment. We estimate each one by a 2×2 DiD of mean outcomes $\bar{Y}$ between cohort g and a comparison group $\mathscr{C}$, from a base period b to t:

$$\widehat{\mathrm{ATT}}(g,t) = (\bar{Y}_{g,t} - \bar{Y}_{g,b}) - (\bar{Y}_{\mathscr{C},t} - \bar{Y}_{\mathscr{C},b}).$$

$\mathscr{C}$ contains the never-treated units (did default) or the units not yet treated in t. For $t \geq g$, $b = g-1$. For $t < g$, $b = t-1$ with the varying base period (did default) and $b = g-1$ with the universal base period (Figure 1). With K periods of anticipation, we use $g-1-K$ in place of $g-1$. Under parallel trends conditional on pre-treatment covariates, we estimate each comparison by outcome regression, inverse probability weighting, or doubly robust estimation; without covariates, the three coincide. For event time ℓ, we average over the set $\mathscr{G}_\ell$ of cohorts observed at ℓ, weighting by cohort size n_g:

$$\hat{\theta}(\ell) = \textstyle\sum_{g \in \mathscr{G}_\ell} w_g(\ell)\, \widehat{\mathrm{ATT}}(g, g+\ell), \quad w_g(\ell) = n_g \big/ \textstyle\sum_{h \in \mathscr{G}_\ell} n_h.$$

In did, `att_gt()` returns $\widehat{\mathrm{ATT}}(g,t)$ and `aggte()` returns $\hat{\theta}(\ell)$ with the dynamic aggregation (default: by cohort).

3. Sun and Abraham

SA estimate every $\mathrm{CATT}_{g,\ell} = \mathrm{ATT}(g, g+\ell)$ in one regression:

$$Y_{it} = \alpha_i + \lambda_t + \textstyle\sum_{g \notin \mathscr{C}} \sum_{\ell \neq -1} \delta_{g,\ell} \mathbf{1}\{G_i = g\} \mathbf{1}\{t - G_i = \ell\} + \varepsilon_{it}.$$

The control cohort $\mathscr{C}$ consists of the never-treated units or, if none exist, the last-treated cohort; SA then drop periods $t \geq \max_i G_i$. With reference period $\ell = -1$ for every cohort, $\hat{\delta}_{g,\ell}$ in a balanced panel equals the CS 2×2 DiD with universal base period $g-1$. The interaction-weighted estimator is $\hat{\nu}_\ell = \sum_{g \in \mathscr{G}_\ell} w_g(\ell)\, \hat{\delta}_{g,\ell}$. SA note that researchers can substitute covariate-adjusted CS estimates for $\hat{\delta}_{g,\ell}$. In fixest, `feols(y ~ sunab(g, t) | id + t)` reports $\hat{\nu}_\ell$ by default.

4. When the point estimates agree

The CS and SA estimates of every $\mathrm{ATT}(g,t)$ and $\theta(\ell)$ are equal when the panel is balanced, there are no covariates, both comparison groups consist of the never-treated units, the CS base period is universal with zero anticipation, and the SA regression has an indicator for every event time except -1 (no binning; the fixest defaults). SA state that the two estimators coincide without covariates and with never-treated units.

Table 1: CS versus SA in the papers, did 2.5.1 (`att_gt()`, `aggte()`), and fixest 0.14.2 (`feols()` with `sunab()`). Point estimates depend on rows 1–5 (row 2: pre-treatment only) and agree when the choices match; rows 6–9 give package defaults for inference. g, t, b, K: cohort, period, base period, anticipation periods; IPW: inverse probability weighting.

Setting	CS (did)	SA (paper; fixest)
1. Comparison group	Never-treated (default) or not-yet-treated units	Never-treated, else last-treated cohort
2. Base period, $t < g$	$b = t-1$ (varying, default) or $g-1$ (universal)	$b = g-1$
3. Covariates	Outcome regression, IPW, or doubly robust (default)	None; in fixest, covariates are regressors
4. No never-treated units	Last-treated cohort; periods before its treatment	Same (in fixest, we drop later periods)
5. Anticipation	$b = g-1-K$ for K periods	We set reference event time $-1-K$
6. Standard errors	Multiplier bootstrap, 1,000 draws	Analytic; homoskedastic errors
7. Clustering	By unit	None (the default since version 0.13.0)
8. 95% intervals	Simultaneous band over all reported estimates	Pointwise
9. Variation in weights	Included in event-study standard errors	Not included (fixed sample shares)

References. Bergé, L. (2026). fixest: Fast Fixed-Effects Estimations. R package version 0.14.2, CRAN. • Callaway, B., and P. H. C. Sant'Anna (2021). Difference-in-Differences with multiple time periods. *Journal of Econometrics* 225(2), 200–230. doi:10.1016/j.jeconom.2020.12.001. • Callaway, B., and P. H. C. Sant'Anna (2026). did: Difference in Differences. R package version 2.5.1, CRAN. • Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics* 225(2), 254–277. doi:10.1016/j.jeconom.2021.03.014. • Sun, L., and S. Abraham (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics* 225(2), 175–199. doi:10.1016/j.jeconom.2020.09.006.